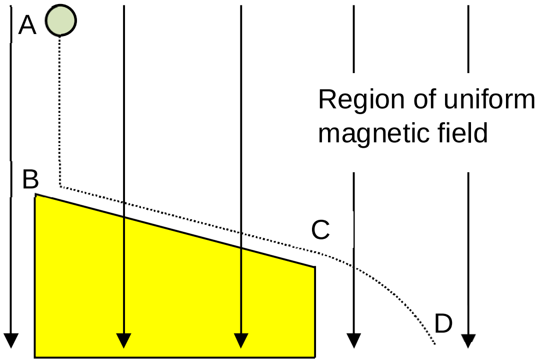
it is an equal action and reaction force.

24 B



$E = Blv$, where v is the horizontal component of the rod's velocity. From point A to B, the rod is moving vertically along the magnetic field, hence E is zero. From B to C, as the rod rolls down the slope, the component of its weight parallel to the slope caused its velocity to increase. Hence, the horizontal velocity component increases at a constant rate and E increases linearly too. From C to D, the rod is moving in projectile motion. Its horizontal velocity component is constant and E remains constant.

25 C The output voltage is 240/20 = 12 V, and output current is 12/5 A